

Mathematical Foundations of Political Methodology

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August 28th, 2017

- Vectors
- Matrix
- Matrix Algebra
- Matrix Inversion

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Points and Vectors

Definition

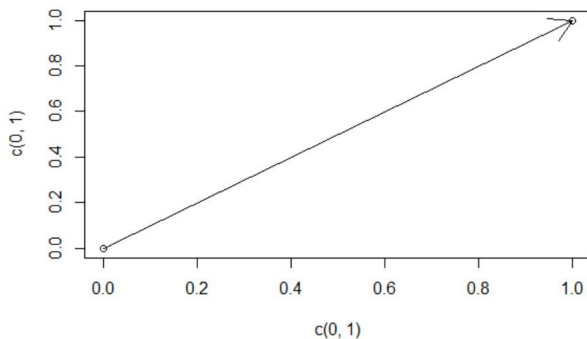
A point $\mathbf{x} \in R^n$ is an ordered n -tuple, $(x_1, x_2, \dots, x_n)$. The vector $\mathbf{x} \in R^n$ is the arrow pointing from the origin $(0, 0, \dots, 0)$ to $\mathbf{x}$.

Some Examples:

- (Latitude, Longitude, Elevation)
- (1, 2, 3)
- (0, 1, 0)

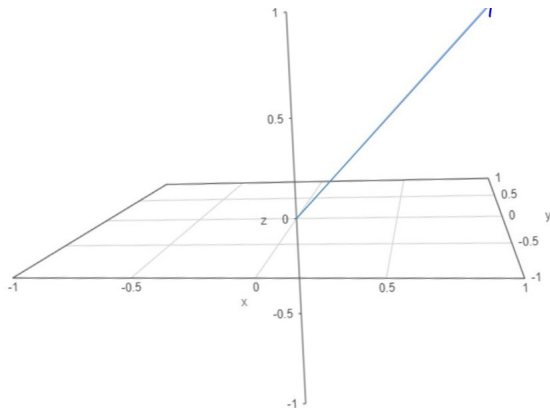
Points and Vectors

Now, let's see a vector in a 2 dimensional space.



Points and Vectors

Now, let's see a vector in a 3 dimensional space.



Arithmetic with Vectors

Definition

Suppose $\mathbf{u}$ and $\mathbf{v}$ are vectors $\mathbf{u}, \mathbf{v} \in R^n$,

$$\mathbf{u} = (u_1, u_2, u_3, \dots, u_n)$$

$$\mathbf{v} = (v_1, v_2, v_3, \dots, v_n)$$

We will say $\mathbf{u} = \mathbf{v}$ if $u_1 = v_1, u_2 = v_2, \dots, u_n = v_n$

Define the sum of $\mathbf{u} + \mathbf{v}$ as

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, u_2 + v_2, u_3 + v_3, \dots, u_n + v_n)$$

Suppose $k \in R$. We will call k a scalar.

Define $k\mathbf{u}$ as the scalar product

$$k\mathbf{u} = (ku_1, ku_2, \dots, ku_n)$$

Examples

Suppose:

$$\mathbf{u} = (1, 2, 3, 4, 5)$$

$$\mathbf{v} = (1, 1, 1, 1, 1)$$

$$k = 2$$

Then,

$$\mathbf{u} + \mathbf{v} = (1 + 1, 2 + 1, 3 + 1, 4 + 1, 5 + 1) = (2, 3, 4, 5, 6)$$

$$k\mathbf{u} = (2 \times 1, 2 \times 2, 2 \times 3, 2 \times 4, 2 \times 5) = (2, 4, 6, 8, 10)$$

$$k\mathbf{v} = (2 \times 1, 2 \times 1, 2 \times 1, 2 \times 1, 2 \times 1) = (2, 2, 2, 2, 2)$$

Vector Lengths

Mathematics calls measures of distance a norm, written $\|\cdot\|$.

The length of vectors in R^n can be measured using the **Euclidean** norm.

Definition

On an n -dimensional Euclidean space R^n , the Euclidean norm of vector, $x = (x_1, x_2, \dots, x_n)$, is defined by the formula

$$\|x\|_2 := \sqrt{x_1^2 + \dots + x_n^2}.$$

An example. What's the Euclidean norm of $x = (1, 3, 5)$?

Its Euclidean norm equals to:

$$\sqrt{1^2 + 3^2 + 5^2} = \sqrt{35}$$

Dot Product

$$x = (x_1, x_2, \dots, x_n)$$

$$y = (y_1, y_2, \dots, y_n)$$

$$x \cdot y = \sum_{i=1}^n x_i y_i = x_1 * y_1 + x_2 * y_2 + \dots + x_n * y_n$$

- Vectors
- **Matrix**
- Matrix Algebra
- Matrix Inversion

Matrices

Matrices collect vectors for convenience.

The first example is the following system of equations:

$$2x + 2y + 3z = 1$$

$$3x + 5y + 7z = 2$$

$$5x + 8y + 3z = 3$$

The coefficient of unknowns can be written compactly as matrices:

$$\begin{bmatrix} 2 & 2 & 3 \\ 3 & 5 & 7 \\ 5 & 8 & 3 \end{bmatrix}$$

Matrices

A second example of matrix is the data set.

	id	name	year	urbanpop	totalpop	urbrate	fpresence	westelbe	protestant	latitude	longitude
1	1	Rhineland	1700	83000	.	.	19	1	.25	50.35	7.6
2	1	Rhineland	1750	129000	1438817	8.9657	19	1	.25	50.35	7.6
3	1	Rhineland	1800	274000	1860719	14.72549	19	1	.25	50.35	7.6
4	1	Rhineland	1850	572278	2846708	20.10317	19	1	.25	50.35	7.6
5	1	Rhineland	1875	1629051	3804381	42.8204	19	1	.25	50.35	7.6
6	1	Rhineland	1880	1850627	4074000	45.42531	19	1	.25	50.35	7.6
7	1	Rhineland	1885	2103753	4344527	48.42306	19	1	.25	50.35	7.6
8	1	Rhineland	1895	2836140	5106002	55.54522	19	1	.25	50.35	7.6
9	1	Rhineland	1900	3506587	5759798	60.88038	19	1	.25	50.35	7.6
10	1	Rhineland	1905	4145633	6436337	64.40982	19	1	.25	50.35	7.6
11	1	Rhineland	1910	4806725	7121140	67.49937	19	1	.25	50.35	7.6
12	2	Mark	1700	5000	117963	4.238618	6	1	.6	51.68333	7.816667
13	2	Mark	1750	5000	149869	3.336254	6	1	.6	51.68333	7.816667
14	2	Mark	1800	36000	193096	18.64353	6	1	.6	51.68333	7.816667
15	2	Mark	1850	55628	362392	15.35011	6	1	.6	51.68333	7.816667
16	2	Mark	1875	305301	721466	42.31675	6	1	.6	51.68333	7.816667
17	2	Mark	1880	.	.	.	6	1	.6	51.68333	7.816667
18	2	Mark	1885	.	.	.	6	1	.6	51.68333	7.816667
19	2	Mark	1895	.	.	.	6	1	.6	51.68333	7.816667
20	2	Mark	1900	1134655	1514546	74.91717	6	1	.6	51.68333	7.816667
21	2	Mark	1905	.	.	.	6	1	.6	51.68333	7.816667
22	2	Mark	1910	.	.	.	6	1	.6	51.68333	7.816667
23	3	Westphalia without Mark	1700	7000	.	.	6	1	.4	51.96667	7.633333
24	3	Westphalia without Mark	1750	14000	528851	2.647246	6	1	.4	51.96667	7.633333
25	3	Westphalia without Mark	1800	55000	763110	7.20735	6	1	.4	51.96667	7.633333
26	3	Westphalia without Mark	1850	61000	1120946	5.441833	6	1	.4	51.96667	7.633333
27	3	Westphalia without Mark	1875	190503	1184231	16.08664	6	1	.4	51.96667	7.633333
28	3	Westphalia without Mark	1880	.	.	.	6	1	.4	51.96667	7.633333

Matrices

Definition

A Matrix is a rectangular array of numbers

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

*If $\mathbf{A}$ has m rows n columns (**R**oman **C**atholic; **R**um & **C**oke) we will say that $\mathbf{A}$ is an $m \times n$ matrix.*

Suppose $\mathbf{X}$ and $\mathbf{Y}$ are $m \times n$ matrices. Then $\mathbf{X} = \mathbf{Y}$ if $x_{ij} = y_{ij}$ for all i and j

Simple Examples

$$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \end{pmatrix}$$

If $\mathbf{I}$ is an $n \times n$ matrix we will call an identity matrix.

Simple Examples

$$\mathbf{X} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \end{pmatrix}$$

$\mathbf{X}$ is an 2×3 matrix

- Vectors
- Matrix
- **Matrix Algebra**
- Matrix Inversion

Matrix Algebra

Definition

Suppose $\mathbf{X}$ and $\mathbf{Y}$ are $m \times n$ matrices. Then define

$$\begin{aligned}\mathbf{X} + \mathbf{Y} &= \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} + \begin{pmatrix} y_{11} & y_{12} & \cdots & y_{1n} \\ y_{21} & y_{22} & \cdots & y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ y_{m1} & y_{m2} & \cdots & y_{mn} \end{pmatrix} \\ &= \begin{pmatrix} x_{11} + y_{11} & x_{12} + y_{12} & \cdots & x_{1n} + y_{1n} \\ x_{21} + y_{21} & x_{22} + y_{22} & \cdots & x_{2n} + y_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} + y_{m1} & x_{m2} + y_{m2} & \cdots & x_{mn} + y_{mn} \end{pmatrix}\end{aligned}$$

Matrix Algebra

Definition

Suppose $\mathbf{X}$ is an $m \times n$ matrix and $k \in R$. Then,

$$k\mathbf{X} = \begin{pmatrix} kx_{11} & kx_{12} & \dots & kx_{1n} \\ kx_{21} & kx_{22} & \dots & kx_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ kx_{m1} & kx_{m2} & \dots & kx_{mn} \end{pmatrix}$$

Matrix Transpose

We will use matrix transpose to flip the dimensionality of a matrix

$$\mathbf{X} = \begin{pmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ x_{21} & x_{22} & \dots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mn} \end{pmatrix}$$
$$\mathbf{X}' = \begin{pmatrix} x_{11} & x_{21} & \dots & x_{m1} \\ x_{12} & x_{22} & \dots & x_{m2} \\ \vdots & \vdots & \ddots & \vdots \\ x_{1n} & x_{2n} & \dots & x_{mn} \end{pmatrix}$$

If $\mathbf{X}$ is an $m \times n$ then $\mathbf{X}'$ is $n \times m$.

If $\mathbf{X} = \mathbf{X}'$ then we say $\mathbf{X}$ is symmetric.

Matrix Transpose

Example 1: $\mathbf{X} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$ then $\mathbf{X}' = \begin{pmatrix} 4 & 1 \\ 1 & 2 \\ 2 & 3 \end{pmatrix}$

Matrix Multiplication

Matrix multiplication is at the root of most econometrics, statistical modeling, but the procedure is not straightforward.

Suppose we have two matrices

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$$

We will create a new matrix $\mathbf{A}$ by matrix multiplication:

$$\begin{aligned} \mathbf{A} &= \mathbf{XY} \\ &= \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \\ &= \begin{pmatrix} 1 \times 1 + 1 \times 3 & 1 \times 2 + 1 \times 4 \\ 1 \times 1 + 2 \times 3 & 1 \times 2 + 2 \times 4 \end{pmatrix} \\ &= \begin{pmatrix} 4 & 6 \\ 7 & 10 \end{pmatrix} \end{aligned}$$

Matrix Multiplication

Definition

Suppose $\mathbf{X}$ is an $m \times n$ matrix and $\mathbf{Y}$ is an $n \times k$ matrix. Then define the matrix $\mathbf{A}$ an $m \times k$ matrix that obtains from multiplying $\mathbf{X}$ and $\mathbf{Y}$ as,

$$\mathbf{A} = \mathbf{XY}$$

$$= \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{pmatrix} \begin{pmatrix} y_{11} & y_{12} & \cdots & y_{1k} \\ y_{21} & y_{22} & \cdots & y_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ y_{n1} & y_{n2} & \cdots & y_{nk} \end{pmatrix}$$

$$= \begin{pmatrix} x_{11}y_{11} + x_{12}y_{21} + \cdots + x_{1n}y_{n1} & \cdots & x_{11}y_{1k} + x_{12}y_{2k} + \cdots + x_{1n}y_{nk} \\ \vdots & \ddots & \vdots \\ x_{m1}y_{11} + x_{m2}y_{21} + \cdots + x_{mn}y_{n1} & \cdots & x_{m1}y_{1k} + x_{m2}y_{2k} + \cdots + x_{mn}y_{nk} \end{pmatrix}$$

Row Vectors

Definition

Suppose $\mathbf{X}$ is an $m \times n$ matrix. We write the row vectors of $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_m \end{pmatrix}$

$$\mathbf{X} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{pmatrix}$$

where

$$\mathbf{x}_1 = (4 \ 1 \ 2)$$

$$\mathbf{x}_2 = (1 \ 2 \ 3)$$

Column Vectors

Definition

Suppose $\mathbf{X}$ is an $m \times n$ matrix. We write the column vectors of $\mathbf{X} = (\mathbf{x}_1 \ \mathbf{x}_2 \ \dots \ \mathbf{x}_m)$

$$\mathbf{X} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}$$

$$\mathbf{X} = (\mathbf{x}_1 \ \mathbf{x}_2 \ \mathbf{x}_3)$$

where

$$\mathbf{x}_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \mathbf{x}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \mathbf{x}_3 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

Definition

Suppose $\mathbf{X}$ is an $m \times n$ matrix and $\mathbf{Y}$ is an $n \times k$ matrix. Write the row

vectors of $\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_m \end{pmatrix}$ and $\mathbf{Y}$ as column vector $\mathbf{Y} = (\mathbf{y}_1 \ \mathbf{y}_2 \ \dots \ \mathbf{y}_k)$.

Then the $m \times k$ matrix $\mathbf{A} = \mathbf{X}\mathbf{Y}$ can be written as

$$\mathbf{A} = \begin{pmatrix} \mathbf{x}_1 \cdot \mathbf{y}_1 & \mathbf{x}_1 \cdot \mathbf{y}_2 & \dots & \mathbf{x}_1 \cdot \mathbf{y}_k \\ \mathbf{x}_2 \cdot \mathbf{y}_1 & \mathbf{x}_2 \cdot \mathbf{y}_2 & \dots & \mathbf{x}_2 \cdot \mathbf{y}_k \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{x}_m \cdot \mathbf{y}_1 & \mathbf{x}_m \cdot \mathbf{y}_2 & \dots & \mathbf{x}_m \cdot \mathbf{y}_k \end{pmatrix}$$

Matrix Multiplication

Let's work on an example together!

$$\mathbf{X} = \begin{pmatrix} 1 & 4 & 5 \\ 10 & 2 & 3 \end{pmatrix}$$

$$\mathbf{Y} = \begin{pmatrix} 2 & 3 \\ 1 & 5 \\ 3 & 5 \end{pmatrix} \text{ What is } \mathbf{XY}?$$

Matrix Multiplication with a Vector

Suppose $\mathbf{X} = \begin{pmatrix} 2 & 3 & 4 & 5 \\ 1 & 5 & 1 & 2 \\ 3 & 5 & 3 & 4 \end{pmatrix}$ a 3×4 matrix and that $\mathbf{v} = \begin{pmatrix} 3 \\ 3 \\ 4 \\ 10 \end{pmatrix}$ a 4×1

matrix (or a column vector) what is

$\mathbf{X}\mathbf{v}$?

What is $\mathbf{X}'\mathbf{v}$?

Algebraic Properties

Suppose $\mathbf{X}$ is an $m \times n$ matrix and $\mathbf{Y}$ is an $n \times k$ matrix. Suppose that

$\mathbf{I} = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$ as the identity matrix and that $k \in R$.

- $\mathbf{XY} \neq \mathbf{YX}$ in general
- $\mathbf{XI} = \mathbf{X}$
- $(\mathbf{X}')' = \mathbf{X}$
- $(\mathbf{XY})' = \mathbf{Y}'\mathbf{X}'$
- $(k\mathbf{X})' = k\mathbf{X}'$
- $(\mathbf{X} + \mathbf{Y})' = \mathbf{X}' + \mathbf{Y}'$

- Vectors
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- Matrix Algebra
- Matrix Inversion

Matrix Inversion: Motivations

Big topic: suppose $\mathbf{X}$ is an $n \times n$ matrix. We want to find the matrix $\mathbf{X}^{-1}$ such that

$$\mathbf{X}^{-1}\mathbf{X} = \mathbf{X}\mathbf{X}^{-1} = \mathbf{I}$$

where $\mathbf{I}$ is the $n \times n$ identity matrix.

Why do we care about matrix inversion?

It will help us to determine whether a system of equations is solvable.

Matrix Inversion: Motivations

Consider the following equations:

$$x_1 + x_2 + x_3 = 0$$

$$0x_1 + 5x_2 + 0x_3 = 5$$

$$0x_1 + 0x_2 + 3x_3 = 6$$

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 5 & 0 \\ 0 & 0 & 3 \end{pmatrix}$$

$$\mathbf{x} = (x_1, x_2, x_3)$$

$$\mathbf{b} = (0, 5, 6)$$

The system of equations are now,

$$\mathbf{Ax} = \mathbf{b}$$

$\mathbf{A}^{-1}$ exists if and only if $\mathbf{Ax} = \mathbf{b}$ has only one solution.

Matrix Inversion, Definition

Definition

Suppose $\mathbf{X}$ is an $n \times n$ matrix. We will call $\mathbf{X}^{-1}$ the inverse of $\mathbf{X}$ if

$$\mathbf{X}^{-1}\mathbf{X} = \mathbf{X}\mathbf{X}^{-1} = \mathbf{I}$$

If $\mathbf{X}^{-1}$ exists then $\mathbf{X}$ is invertible. If $\mathbf{X}^{-1}$ does not exist, then we will say $\mathbf{X}$ is singular.

Matrix Inversion

Now, let's discuss some properties of matrix inversion.

Theorem

The inverse of matrix $\mathbf{X}$, $\mathbf{X}^{-1}$, is unique

Proof.

By way of contradiction, suppose not. Then there are at least two matrices $\mathbf{A}$ and $\mathbf{C}$ such that $\mathbf{AX} = \mathbf{I}$ and $\mathbf{CX} = \mathbf{I}$

This implies that,

$$\begin{aligned}\mathbf{AXC} &= (\mathbf{AX})\mathbf{C} \\ &= \mathbf{IC} \\ &= \mathbf{C}\end{aligned}$$

Matrix Inversion

But it also implies that

$$\begin{aligned}\mathbf{AXC} &= \mathbf{A(XC)} \\ &= \mathbf{A(I)} \\ &= \mathbf{A}\end{aligned}$$

So $\mathbf{C} = \mathbf{AXC} = \mathbf{A}$ or $\mathbf{C} = \mathbf{A}$ but this contradicts our assumption that there are two unique inverses.

Matrix Inversion

Theorem

Suppose $\mathbf{A}$ has inverse $\mathbf{A}^{-1}$ and $\mathbf{B}$ has inverse $\mathbf{B}^{-1}$. Then,

$$(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

Proof.

We need to show that $(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) = (\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}$.

$$\begin{aligned}(\mathbf{B}^{-1}\mathbf{A}^{-1})(\mathbf{AB}) &= \mathbf{B}^{-1}(\mathbf{A}^{-1}\mathbf{A})\mathbf{B} \\&= \mathbf{B}^{-1}\mathbf{I}\mathbf{B} \\&= \mathbf{B}^{-1}\mathbf{B} \\&= \mathbf{I}\end{aligned}$$

Matrix Inversion

$$\begin{aligned}(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) &= \mathbf{A}(\mathbf{BB}^{-1})\mathbf{A}^{-1} \\ &= \mathbf{A}\mathbf{I}\mathbf{A}^{-1} \\ &= \mathbf{AA}^{-1} \\ &= \mathbf{I}\end{aligned}$$

So $\mathbf{AB}$ is invertible and $(\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$.

Inclass Exercise: Challenge Inversion Proofs

- Show that $(\mathbf{A}^{-1})^{-1} = \mathbf{A}$.
- Show that $(k\mathbf{A})^{-1} = \frac{1}{k}\mathbf{A}^{-1}$

Matrix Inversion

After we define matrix inversion and discuss its properties, a natural question to ask is that when do matrix inversion exist?

The answer relates to a concept, linear independence.

Intuitively, linear independence means that there is no repeated information in the matrix.

Matrix Inversion: Existence

Definition

Suppose we have a set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_r\}$

And consider the system of equations

$$k_1 \mathbf{v}_1 + k_2 \mathbf{v}_2 + \dots + k_r \mathbf{v}_r = \mathbf{0}$$

If the only solution is $k_1 = 0, k_2 = 0, k_3 = 0, \dots, k_r = 0$ then we say that the set is linearly independent. If there are other solutions, then the set is linearly dependent.

Matrix Inversion: Existence

Consider $\mathbf{v}_1 = (1, 0, 0)$, $\mathbf{v}_2 = (0, 1, 0)$, $\mathbf{v}_3 = (0, 0, 1)$

Can we write this as a combination of other vectors?

Consider $\mathbf{v}_1 = (1, 0, 0)$, $\mathbf{v}_2 = (0, 1, 0)$, $\mathbf{v}_3 = (0, 0, 1)$, $\mathbf{v}_4 = (1, 2, 3)$.

Can we write any of these vector as a combination of the remaining two vectors?

$$\mathbf{v}_4 = \mathbf{v}_1 + 2\mathbf{v}_2 + 3\mathbf{v}_3$$

Matrix Inversion: Existence

Theorem

Suppose $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_K \in R^n$. If $K > n$ then the set is linearly dependent

- $\mathbf{v}_1 = (v_{11}, v_{21}, \dots, v_{n1})$
- Says that if there are more vectors in the set than elements in each vector, one must be linearly dependent

Matrix Inversion: Existence

The following theorem tells that

Theorem

Suppose $\mathbf{X}$ is an $n \times n$ matrix. Recall we can write this matrix as $\begin{pmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{pmatrix}$.

Then $\mathbf{X}$ has an inverse if and only if $S = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ is linearly independent

If this is true, we say $\mathbf{X}$ has full rank.

Matrix Inversion: Rank and Span

$$\mathbf{A} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

Columns 1 and 2 are linearly independent, but their sum equals Column 3; the rank of $\mathbf{A} = 2$

These three vectors span $\mathbb{R}^2$ but not $\mathbb{R}^3$: Show that for any $(x_1, x_2) \in \mathbb{R}^2$, we can find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1, 0, 1) + \lambda_2(0, 1, 1) + \lambda_3(1, 1, 2) = (x_1, x_2, x_1 + x_2)$$

They span $\mathbb{R}^2$ and the space of all vectors in $\mathbb{R}^3$ of form $(x_1, x_2, x_1 + x_2)$

Show that for any $(x_1, x_2, x_3) \in \mathbb{R}^3$, we *can't* find a $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(1, 0, 1) + \lambda_2(0, 1, 1) + \lambda_3(1, 1, 2) = (x_1, x_2, x_3)$$

Matrix Inversion: Rank and Span

Finding the rank of a matrix $\mathbf{A}$

- Reduce $\mathbf{A}$ to an echelon matrix $\mathbf{E}$ using Gaussian elimination
- The number of non-zero rows of $\mathbf{E}$ is the rank of $\mathbf{A}$!

Properties of ranks

- The rank of an $n \times m$ matrix $\mathbf{A}$ can't be greater than either n or m :
 $\text{Rank}(\mathbf{A}) \leq \min(n, m)$
- A matrix has *full rank* if its rank is the biggest it could be:
 $\text{Rank}(\mathbf{A}) = \min(n, m)$
- A square matrix is *invertible* if and only if it has full rank

Matrix Inversion: Gauss-Jordan elimination

How to invert a matrix

Take

$$[\mathbf{A} \mathbf{I}]$$

and use Gauss-Jordan elimination to convert it into

$$[\mathbf{I} \mathbf{A}^{-1}]$$

Gauss-Jordan elimination is similar to Gaussian elimination, but turns matrix into a diagonal matrix, instead of just an echelon matrix

Matrix Inversion: Gauss-Jordan elimination

Let's invert

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 2 \\ 3 & 1 & 1 \\ 3 & 1 & 2 \end{bmatrix}$$

$$\text{Step 1: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 0 \\ 3 & 1 & 2 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{Step 2: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 2: } \begin{bmatrix} 2 & 1 & 2 & 1 & 0 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & -\frac{1}{2} & -1 & -\frac{3}{2} & 0 & 1 \end{bmatrix}$$

Since we are creating a diagonal we want to get rid of the “1” in (1, 2)

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

This added $2 \times \text{Row 2}$ to Row 1 and subtracted Row 2 from Row 3

Matrix Inversion: Gauss-Jordan elimination

$$\text{Step 3: } \begin{bmatrix} 2 & 0 & -2 & -2 & 2 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{3}{2} & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We get a diagonal matrix by adding $2 \times \text{Row 3}$ to Rows 1 & 2

$$\text{Step 4: } \begin{bmatrix} 2 & 0 & 0 & -2 & 0 & 2 \\ 0 & -\frac{1}{2} & 0 & -\frac{3}{2} & -1 & 2 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix}$$

We finish by normalizing

$$\text{Step 5: } \begin{bmatrix} 1 & 0 & 0 & -1 & 0 & 1 \\ 0 & 1 & 0 & 3 & 2 & -4 \\ 0 & 0 & 1 & 0 & -1 & 1 \end{bmatrix} \text{ and } \mathbf{A}^{-1} = \begin{bmatrix} -1 & 0 & 1 \\ 3 & 2 & -4 \\ 0 & -1 & 1 \end{bmatrix}$$

Matrix Inversion: An Application

Suppose we have following system of equations:

$$2x + 1y + 2z = 1$$

$$3x + 1y + 1z = 3$$

$$3x + 1y + 2z = 5$$

Say, $\mathbf{x} = (x, y, z)'$, $\mathbf{b} = (1, 3, 5)'$.

Say, the coefficient matrix for unknowns is A , the inverse of A is computed by the previous example.

Then, we can solve the unknowns by:

$$\mathbf{x} = A^{-1}\mathbf{b} = (4, -11, 2)'$$